

PROJECT REPORT

DEMAND FORECASTING AND INVENTORY OPTIMIZATION USING MACHINE LEARNING

A Project Report Submitted in Partial Fulfillment of the Requirements for
the Business Analytics Course

Submitted by

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ABSTRACT

This study presents a machine learning-based framework for demand forecasting and inventory optimization using historical retail sales data, where SARIMA, Random Forest, and XGBoost models were developed and compared, with XGBoost achieving the best forecasting performance. The forecasted demand was integrated with Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP) models to support efficient inventory planning, while an interactive Tableau dashboard was developed to visualize forecasting results and inventory insights for data-driven decision-making.

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Chapter 1: INTRODUCTION

1.1 Introduction

Demand forecasting and inventory optimization are two of the most important functions in modern supply chain and inventory management. Every business, whether it is a retail store, supermarket, manufacturing company, or e-commerce platform, must maintain the right quantity of products to satisfy customer demand while minimizing inventory-related costs. Maintaining this balance is challenging because customer demand is dynamic and influenced by several factors such as seasonality, holidays, promotions, economic conditions, market trends, weather, and consumer behavior.

Businesses generate large volumes of historical sales data through their daily operations. This data contains valuable information about customer purchasing patterns, product demand, seasonal fluctuations, and sales trends. By analyzing historical sales data along with other influencing factors, organizations can gain insights into future demand and make informed inventory decisions.

Demand forecasting is the process of estimating future customer demand based on historical data and various influencing factors. Accurate demand forecasts enable businesses to plan production, procurement, and inventory levels more effectively. Traditionally, statistical time series methods such as Moving Average, Exponential Smoothing, and SARIMA have been widely used for forecasting because they analyze historical demand patterns such as trend and seasonality. However, these methods often struggle to capture complex nonlinear relationships and multiple influencing variables.

Recent advances in Machine Learning have significantly improved forecasting capabilities. Machine learning algorithms can automatically identify hidden patterns, interactions, and nonlinear relationships within large datasets, leading to more accurate demand predictions. Models such as Random Forest and XGBoost can utilize multiple features simultaneously, including product information, store characteristics, seasonal effects, promotions, holidays, and lagged sales variables, thereby providing more robust demand forecasts.

Demand forecasting alone is not sufficient for efficient inventory management. The predicted demand must be translated into inventory decisions that determine how much inventory should be ordered and when replenishment should occur. This process is known as inventory optimization. Inventory optimization aims to maintain sufficient stock to satisfy customer demand while minimizing inventory-related costs such as purchasing cost, ordering cost, holding cost, and shortage cost.

Several inventory management techniques support this objective. Economic Order Quantity (EOQ) determines the optimal order quantity that minimizes the combined ordering and holding costs. Safety Stock provides a buffer against demand uncertainty and supply delays, reducing the risk of stockouts. Reorder Point (ROP) identifies the inventory level at which a new order

should be placed, considering forecasted demand during lead time and safety stock requirements.

This project integrates machine learning-based demand forecasting with inventory optimization techniques to develop a data-driven inventory management framework. By accurately forecasting future demand and calculating optimal inventory parameters such as EOQ, Safety Stock, and Reorder Point, businesses can improve inventory availability, reduce operational costs, enhance customer satisfaction, and support more efficient supply chain decision-making .

1.2 Background and Need of the Study

Inventory management has become increasingly complex due to changing customer preferences, fluctuating market conditions, seasonal demand, promotional activities, and supply chain uncertainties. Businesses frequently face the challenge of maintaining the right inventory levels to satisfy customer demand without incurring unnecessary costs.

When inventory levels are lower than actual demand (understocking), businesses experience stockouts, lost sales, reduced customer satisfaction, and potential loss of loyal customers. Customers may switch to competitors if products are unavailable, negatively affecting both revenue and brand reputation.

Conversely, maintaining excessive inventory (overstocking) results in higher holding costs, increased warehouse utilization, capital being locked in unsold inventory, product deterioration or expiration, obsolescence, and reduced cash flow. These issues significantly impact the financial performance of an organization.

Traditional inventory planning often relies on manual estimation, simple historical averages, or statistical forecasting methods. While these approaches provide useful insights, they may not effectively capture complex demand patterns influenced by multiple variables.

Machine learning provides an opportunity to improve forecasting accuracy by learning hidden relationships from historical sales data and multiple influencing factors. Accurate demand forecasts allow businesses to estimate future product requirements more reliably, enabling inventory optimization techniques such as Economic Order Quantity (EOQ), Safety Stock, and Reorder Point (ROP) to determine the optimal inventory levels.

The integration of machine learning and inventory optimization helps organizations reduce inventory costs, improve product availability, minimize stockouts and overstock situations, and support better business decisions. Therefore, developing an intelligent demand forecasting and inventory optimization framework has become an important area of research for modern supply chain management.

1.3 Problem Statement

Many organizations still rely on traditional inventory management practices that estimate future demand using simple historical averages or conventional statistical methods. Due to fluctuations in customer demand caused by seasonality, promotions, holidays, economic conditions, and other external factors, these approaches often fail to provide sufficiently accurate forecasts

Inaccurate demand forecasting leads to two major inventory problems:

- Understocking, resulting in stockouts, lost sales, delayed customer orders, and reduced customer satisfaction.
- Overstocking, leading to increased holding costs, excess inventory, product expiration or obsolescence, and unnecessary capital investment.

Moreover, businesses often lack an integrated system that combines accurate demand forecasting with inventory optimization techniques to determine optimal ordering quantities and replenishment timing.

Therefore, there is a need for a data-driven framework that utilizes machine learning techniques to improve demand forecasting accuracy and integrates inventory optimization methods such as Economic Order Quantity (EOQ), Safety Stock, and Reorder Point (ROP) to support efficient inventory management and cost reduction.

1.4 Objectives of the Study

To develop a machine learning-based demand forecasting model integrated with inventory optimization techniques for improving inventory management and supporting effective business decision-making.

- To analyze historical sales data and identify demand patterns and perform exploratory data analysis and feature engineering for demand forecasting.
- To develop and compare multiple forecasting models, including Random Forest, XGBoost, and SARIMA and evaluate forecasting performance using metrics such as RMSE, MSE and WAPE
- To select the best-performing forecasting model based on prediction accuracy to estimate future product demand using the selected forecasting model.
- To calculate Economic Order Quantity (EOQ), Safety Stock, and Reorder Point (ROP) using the predicted demand and demonstrate how demand forecasting can support efficient inventory optimization and reduce inventory-related costs.

1.5 Scope of the Study

The scope of this study focuses on integrating demand forecasting with inventory optimization using historical sales data. The study analyzes product-level sales across different stores to

understand demand behavior and generate future demand predictions using both traditional time series methods and machine learning algorithms.

The research includes data preprocessing, exploratory data analysis, feature engineering, model development, performance evaluation, and inventory optimization calculations. Inventory decisions are supported through Economic Order Quantity (EOQ), Safety Stock, and Reorder Point (ROP) using forecasted demand and assumed inventory cost parameters.

The study is limited to demand prediction based on historical sales records and selected business assumptions. Factors such as supplier disruptions, real-time market changes, competitor pricing, customer sentiment, and unexpected external events are beyond the scope of this research. Nevertheless, the proposed framework provides a practical and scalable approach that organizations can adapt to improve inventory planning and supply chain efficiency.

2. LITERATURE REVIEW

2.1 Literature Review on Demand Forecasting

Literature 1

Kumar et al. (2022) proposed a hybrid demand forecasting model by combining Random Forest, XGBoost, and Linear Regression using retail sales and external variables. The hybrid model achieved higher forecasting accuracy than individual models by capturing nonlinear and seasonal demand patterns. However, the study focused only on forecasting and did not address inventory optimization. This supports the use of ensemble machine learning models in the present study. [11]

Literature 2

Kavitha et al. (2023) applied the XGBoost algorithm for retail demand forecasting using historical sales data. The model effectively captured seasonal demand patterns and outperformed conventional statistical methods. However, inventory optimization techniques were not included. The findings support the use of XGBoost in this research. [12]

Literature 3

Ganguly and Mukherjee (2024) compared several machine learning algorithms for retail sales forecasting and found that the optimized Random Forest model produced the highest prediction accuracy. The study focused only on forecasting performance without inventory planning. It validates the effectiveness of tree-based machine learning models for demand prediction. [13]

Literature 4

Shakir and Modupe (2023) evaluated eighteen regression-based machine learning models for demand forecasting. Ensemble models such as Random Forest and XGBoost consistently outperformed traditional regression methods. However, inventory optimization was not considered. The study supports the use of multiple machine learning models in this research. [14]

Literature 5

Bai (2024) compared ARIMA and XGBoost for retail demand forecasting using a public retail dataset. XGBoost generated more accurate forecasts by capturing nonlinear demand patterns, whereas ARIMA mainly modeled linear seasonality. The study provides a basis for comparing statistical and machine learning forecasting methods. [15]

2.2 Literature Review on Inventory Optimization

Literature 6

Jaber and Peltokorpi (2024) reviewed the development of EOQ and EPQ inventory models and confirmed that EOQ remains an effective technique for minimizing ordering and holding costs. However, the study did not integrate machine learning-based demand forecasting. This supports the application of EOQ in the present study. [16]

Literature 7

Fetene et al. (2021) reviewed safety stock determination under demand and supply uncertainty. The authors concluded that combining demand forecasting with safety stock calculations improves inventory performance and reduces stockouts. The review also noted that many studies assume inventory parameters due to limited operational data. [17]

2.3 Literature Review on Integrated Demand Forecasting and Inventory Optimization and the assumption of costs and lead time

Literature 8

Li et al. (2020) proposed an integrated framework that combines machine learning-based demand forecasting with inventory replenishment decisions. The approach reduced stock shortages and ordering frequency while maintaining service levels. Although developed for healthcare inventory, it demonstrates the benefits of integrating forecasting with inventory optimization. [18]

Literature 9

Bergsma, de Ruijt, and Bhulai (2025) systematically reviewed machine learning applications in inventory optimization. The study found that integrating demand forecasting with classical inventory models improves replenishment decisions. It also reported that many public datasets lack inventory cost parameters, requiring researchers to assume inventory costs and lead times. Analyzed 122 studies on machine learning-based inventory control and identified the lack of real industrial inventory datasets as a major research challenge. The review justified the use of assumed inventory parameters for implementing EOQ, Safety Stock, and Reorder Point models when operational cost data are unavailable. [19]

Chapter 3: RESEARCH METHODOLOGY

This study follows a quantitative, data-driven research methodology to develop an integrated demand forecasting and inventory optimization framework. Historical retail sales data obtained from the Kaggle Demand Forecasting dataset were analyzed to identify demand patterns and forecast future sales. The research methodology consists of data collection, preprocessing, exploratory data analysis, feature engineering, forecasting model development, model evaluation, and inventory optimization. Three forecasting models, namely SARIMA, Random Forest, and XGBoost, were developed and compared using multiple evaluation metrics. The best-performing forecasting model was subsequently integrated with inventory optimization techniques, including Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP), to generate practical inventory policies. Finally, Tableau dashboards were developed to visualize forecasting results and inventory recommendations for business decision-making.

3.1 Overall Workflow

The overall workflow of the proposed framework begins with collecting historical sales data from the Kaggle Demand Forecasting dataset. The collected data undergo preprocessing to improve data quality, followed by exploratory data analysis to understand demand trends, seasonality, and sales patterns. Feature engineering techniques are then applied to create additional predictive variables. The processed dataset is divided into training and testing subsets using a chronological train-test split. Three forecasting models—SARIMA, Random Forest, and XGBoost—are trained and evaluated using standard forecasting performance metrics. The best-performing model is selected to predict future demand, which is subsequently used to

calculate Economic Order Quantity (EOQ), Safety Stock, and Reorder Point.

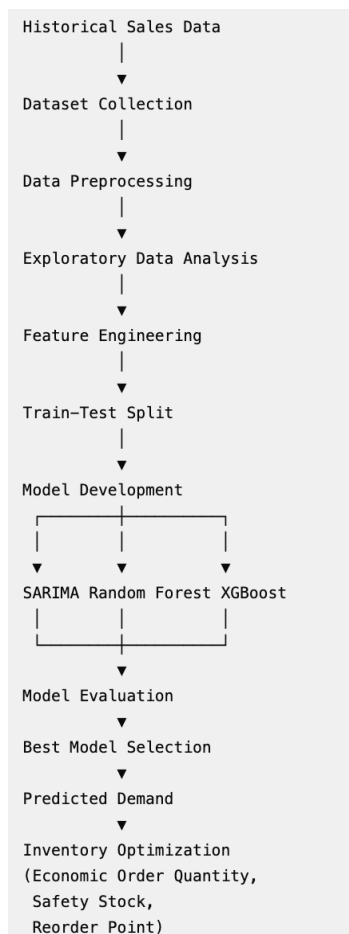


Figure.3.1: Overall Research Workflow

3.2 System Configuration

The experiments were conducted on a personal computer equipped with Apple M2 processor, 16 GB RAM, and 512 GB SSD running macOS. Python served as the primary programming language for implementing data preprocessing, feature engineering, forecasting models, and inventory optimization. The hardware and software configuration was sufficient to efficiently execute all stages of the proposed framework.

3.3 Tools and Technologies

The implementation of the proposed framework utilized several open-source software tools and Python libraries. Python was used as the primary programming language, while Jupyter Notebook and Google Colaboratory provided an interactive development environment. Pandas and NumPy were employed for data manipulation and numerical computation, Matplotlib and Plotly for visualization, Statsmodels for SARIMA implementation, Scikit-learn for Random

Forest, and XGBoost library for gradient boosting. Tableau was used to develop interactive dashboards for visualizing forecasting and inventory optimization results.

The complete implementation of the proposed framework has been maintained using Git for version control and is publicly available through a GitHub repository (see **Appendix A**).

3.4 Dataset Description

The study utilizes the Kaggle Demand Forecasting dataset, which contains historical daily sales records of multiple products sold across different retail stores. The dataset includes four attributes: Date, Store, Item, and Sales. These historical sales records provide valuable information for identifying demand trends, seasonal patterns, and temporal dependencies required for demand forecasting. The dataset was selected because it represents a realistic retail environment and is widely used for forecasting research.

3.5 Data Collection

The dataset was obtained from the publicly available Kaggle Demand Forecasting competition. Since the dataset contains only historical sales information, operational inventory parameters such as ordering cost, holding cost, lead time, and purchase cost were unavailable. Therefore, these parameters were assumed based on established inventory management literature to facilitate inventory optimization analysis..

3.6 Data Preprocessing

Data preprocessing was performed to improve data quality before model development. Missing values, duplicate records, and inconsistent data types were examined. The Date attribute was converted into datetime format to support chronological analysis and feature extraction. To reduce computational complexity and focus on products with the highest business impact, the five products with the highest cumulative sales were selected for detailed forecasting and inventory optimization.

3.7 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the dataset and identify demand behaviour. Sales trend analysis, product-wise and store-wise demand analysis, seasonal decomposition, stationarity testing using the Augmented Dickey-Fuller test, and autocorrelation analysis using ACF and PACF were performed. These analyses provided insights into temporal demand patterns and guided forecasting model selection.

3.8 Feature Engineering

Feature engineering was performed to improve forecasting performance by creating additional variables from the original dataset. Calendar features, lag variables, rolling mean, rolling standard deviation, exponentially weighted moving averages (EWMA), and group-based statistical features were generated to capture temporal dependencies and demand behaviour. These engineered features enhanced the learning capability of machine learning models.

3.9 Model Development

Three forecasting models were developed and compared: SARIMA, Random Forest, and XGBoost. A chronological train-test split was adopted to preserve temporal order and avoid data leakage. SARIMA was implemented as the statistical baseline model, while Random Forest and XGBoost utilized engineered features to capture nonlinear relationships in retail demand. All models were trained using identical training and testing periods to ensure a fair performance comparison.

3.9.1 Train-Test Split

Since demand forecasting is a time series prediction problem, the dataset was divided using a chronological train-test split rather than random sampling. Random splitting would allow future observations to appear in the training data, leading to data leakage and unrealistic forecasting performance. Therefore, the historical sales records were ordered by date, and the first **80%** of the observations were used for model training, while the remaining **20%** were reserved for testing.

The training dataset was used to learn historical demand patterns, trends, and seasonality, whereas the testing dataset was used to evaluate each model's ability to predict unseen future demand. Maintaining the temporal sequence ensures that the forecasting models simulate real-world business scenarios where future demand is predicted solely from past information. This methodology provides a reliable assessment of forecasting accuracy and supports an unbiased comparison between SARIMA, Random Forest, and XGBoost models.

Train-Test Split Configuration

Dataset	Percentage	Purpose
Training Set	80%	Model training and learning historical demand patterns

Testing Set	20%	Model evaluation using unseen future demand
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3.9.2 SARIMA

The SARIMA model was developed after confirming stationarity using the Augmented Dickey-Fuller (ADF) test and selecting appropriate autoregressive and moving average parameters through Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) analysis. SARIMA captures trend, seasonality, and autocorrelation in historical demand data, making it a suitable statistical baseline model for comparison.

3.9.3 Random Forest

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees using bootstrap sampling and random feature selection. In this study, the model was trained using engineered features, including calendar variables, lag features, rolling statistics, and store-item information. The final forecast was obtained by averaging the predictions from all decision trees, improving forecasting stability and reducing overfitting.

3.9.4 XGBoost

XGBoost is a gradient boosting algorithm that builds decision trees sequentially by minimizing prediction errors from previous iterations. It effectively captures complex nonlinear relationships and interactions among engineered features, resulting in highly accurate demand forecasts. Due to its regularization techniques and efficient tree construction, XGBoost has become one of the most widely used machine learning algorithms for demand forecasting applications.

3.10 Model Evaluation

The forecasting models were evaluated using Mean Absolute Error (MAE), which measures the average magnitude of prediction errors without considering their direction. Lower MAE value indicates better forecasting performance.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of observations

Root Mean Squared Error (RMSE), represents the prediction error in the same unit as the original data.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of observations

and Weighted Absolute Percentage Error (WAPE).

$$\text{WAPE} = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{\sum_{i=1}^n y_i} \times 100$$

where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value

These evaluation metrics provided a comprehensive assessment of prediction accuracy and facilitated the selection of the best-performing forecasting model.

3.11 Inventory Optimization Methodology

The predicted demand generated by the selected forecasting model was integrated with classical inventory management techniques to determine optimal inventory policies. Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP) were calculated using forecasted demand together with assumed inventory parameters obtained from inventory management literature.

3.11.1 Assumptions for Inventory Optimization

The Kaggle Demand Forecasting dataset contains only historical sales data and does not include inventory parameters such as ordering cost, holding cost, purchase cost, lead time, or service level, which are required for inventory optimization. Therefore, representative values commonly adopted in inventory management literature were assumed to demonstrate the integration of demand forecasting with Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP). These assumptions provide a realistic basis for evaluating the proposed framework and can be replaced with actual organizational data in real-world applications.

Parameter	Assumed Value	Unit
Ordering Cost	₹100	per order
Holding Cost	₹0.20%	per unit/year
Purchase Cost	₹100	per unit
Lead Time	7 days	days
Service Level	95%	-
Z-value	1.65	-

Table 3.1 Assumed Inventory Parameters

3.11.2 Economic Order Quantity (EOQ)

The Economic Order Quantity (EOQ) model was used to determine the optimal order quantity that minimizes the combined ordering and inventory holding costs. The EOQ model assumes relatively stable demand, constant lead time, and instantaneous replenishment, making it an appropriate baseline inventory model for forecasting-based inventory optimization [12],[14].

The EOQ was calculated using Equation (12.1):

$$EOQ = \sqrt{\frac{2DS}{H}}$$

where:

- **D** = Annual demand (forecasted demand)
- **S** = Ordering cost per order
- **H** = Annual holding cost per unit

3.11.3 Safety Stock

Demand uncertainty and supplier lead-time variability may result in stock shortages if inventory is replenished only after existing stock is exhausted. Therefore, Safety Stock (SS) was calculated to maintain an adequate inventory buffer against unexpected demand fluctuations.

Safety Stock was computed using the following equation:

$$SS = Z \times \sigma_d \times \sqrt{L}$$

where:

- **Z** = Service level factor
- **σ_d** = Standard deviation of demand
- **L** = Lead time

A **95% service level** was assumed in this study, corresponding to a Z-value of approximately **1.65**. [13].

3.11.4 Reorder Point (ROP)

The Reorder Point (ROP) indicates the inventory level at which a new purchase order should be placed to avoid stock shortages during supplier lead time. The ROP was calculated using the forecasted average daily demand together with the calculated safety stock.

The ROP is given by:

$$\text{ROP} = dL + \text{SS}$$

The reorder point was computed using the forecasted average daily demand, assumed lead time, and calculated safety stock, thereby linking predictive analytics with operational inventory control [12].

CHAPTER 4: RESULTS AND DISCUSSION

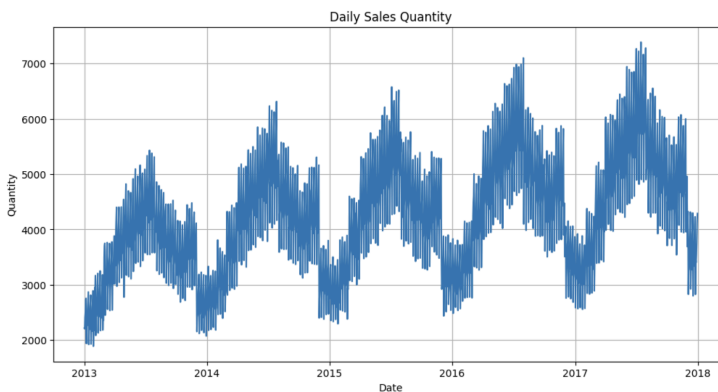
This chapter presents the findings obtained from the exploratory data analysis, demand forecasting models, inventory optimization techniques, and dashboard visualization. The performance of the three forecasting models—SARIMA, Random Forest, and XGBoost—is evaluated and compared using multiple forecasting accuracy metrics. The chapter also discusses the inventory optimization results obtained using Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP), followed by business implications and study limitations.

4.1 Exploratory Data Analysis Results

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the sales dataset and identify demand patterns that influence forecasting performance. The analysis included sales trend analysis, product-wise and store-wise sales comparison, seasonal decomposition, stationarity testing, and autocorrelation analysis.

4.1.1 Sales Trend Analysis

The overall daily sales trend revealed a clear temporal pattern with recurring fluctuations over time. Sales exhibited both long-term growth and periodic seasonal variations, indicating that customer demand changes systematically across different periods. Several demand peaks were observed, suggesting the influence of seasonal events, holidays, or promotional activities.



Insert Figure 4.1: Daily Sales Trend

4.1.2 Product-wise Analysis

Product-wise analysis showed significant variation in demand among products. Certain products consistently generated higher sales volumes than others, indicating differences in customer preferences and purchasing behavior. The five highest-selling products were selected for detailed forecasting and inventory optimization because they contribute substantially to overall sales.

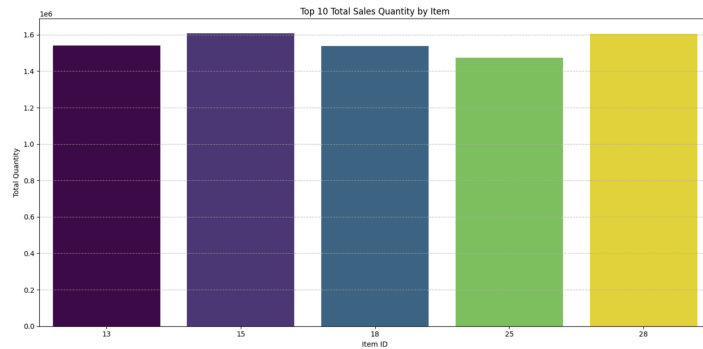


Figure 4.2: Top Five Products Based on Total Sales

4.1.3 Store-wise Analysis

Store-wise sales analysis indicated relatively balanced demand across stores, although moderate differences existed due to varying customer traffic and regional demand patterns. These variations justify the inclusion of store information as an explanatory variable in the machine learning models.

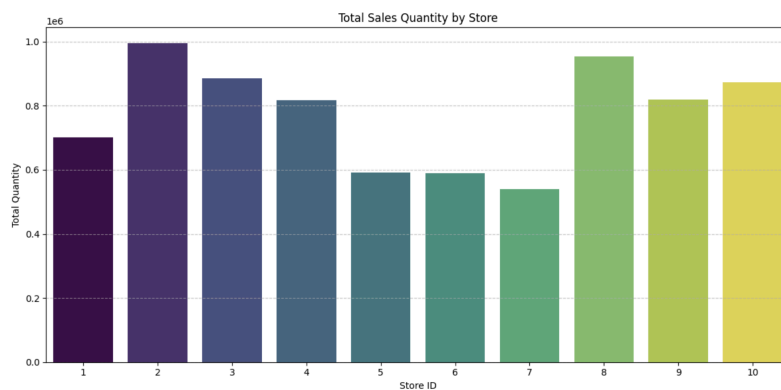


Figure 4.3: Store-wise Sales Distribution

4.1.4 Seasonal Decomposition

Seasonal decomposition separated the time series into trend, seasonal, and residual components. The results confirmed the presence of a stable seasonal pattern together with a gradually increasing long-term trend. The residual component contained random fluctuations that could not be explained solely by trend or seasonality.

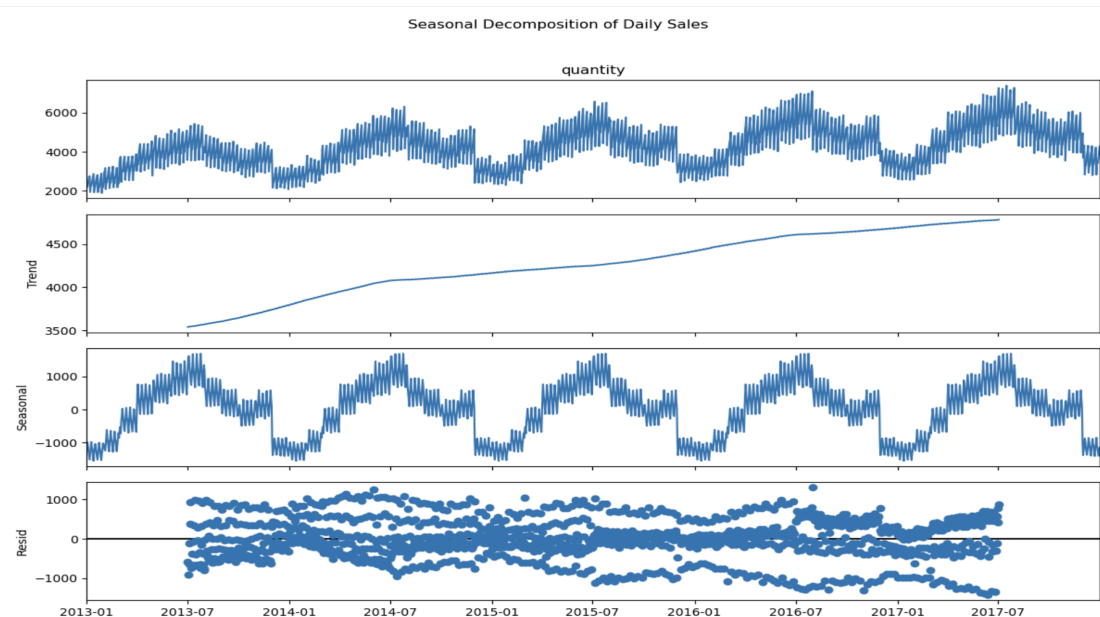


Figure 4.4: Seasonal Decomposition Plot

4.1.5 Augmented Dickey-Fuller (ADF) Test

The Augmented Dickey-Fuller (ADF) test was performed to determine whether the sales time series was stationary. The test results indicated that the series was stationary at the initial stage, as the ADF test statistic was sufficiently negative and the corresponding p-value was less than the significance level of 0.05. Therefore, the null hypothesis of the presence of a unit root was rejected, confirming that the time series was stationary.

Since the data was already stationary, no additional differencing was required before developing the SARIMA model. The stationary nature of the series ensured that the underlying statistical assumptions for time series modeling were satisfied, enabling reliable estimation of model parameters.

```

Performing Augmented Dickey-Fuller Test on Daily Sales...
ADF Statistic: -2.996280
p-value: 0.035243
Lags Used: 21
Number of Observations Used: 1804
Critical Values:
  1%: -3.434
  5%: -2.863
 10%: -2.568
Conclusion: The p-value is less than or equal to 0.05. We reject the null hypothesis, meaning the time series is likely stationary.

```

Table 4.1: ADF Test Results

4.1.6 ACF and PACF Analysis

Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were analyzed to determine appropriate autoregressive and moving average parameters for the SARIMA model. The ACF plot showed gradually decaying correlations, while the PACF plot identified significant lag relationships useful for model parameter selection.

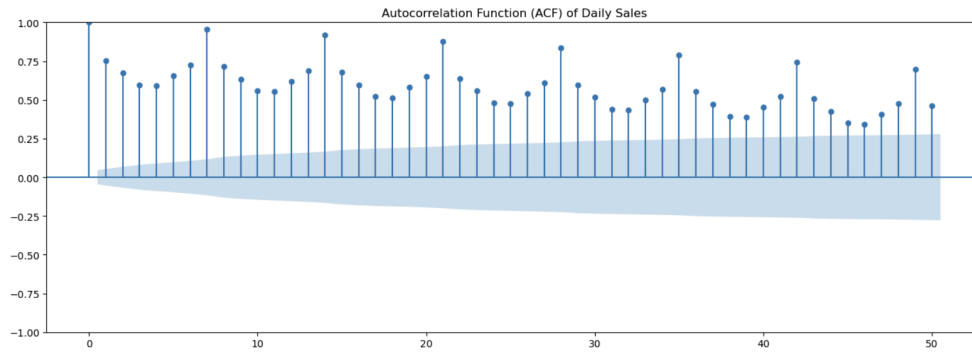


Figure 4.5: ACF Plot

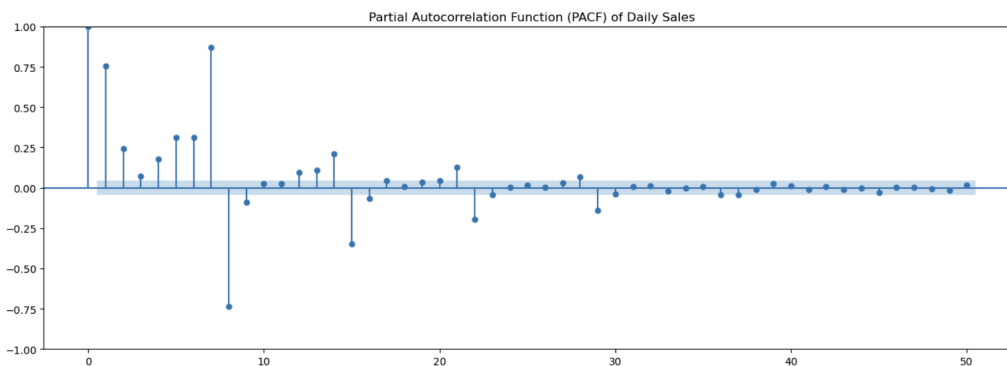


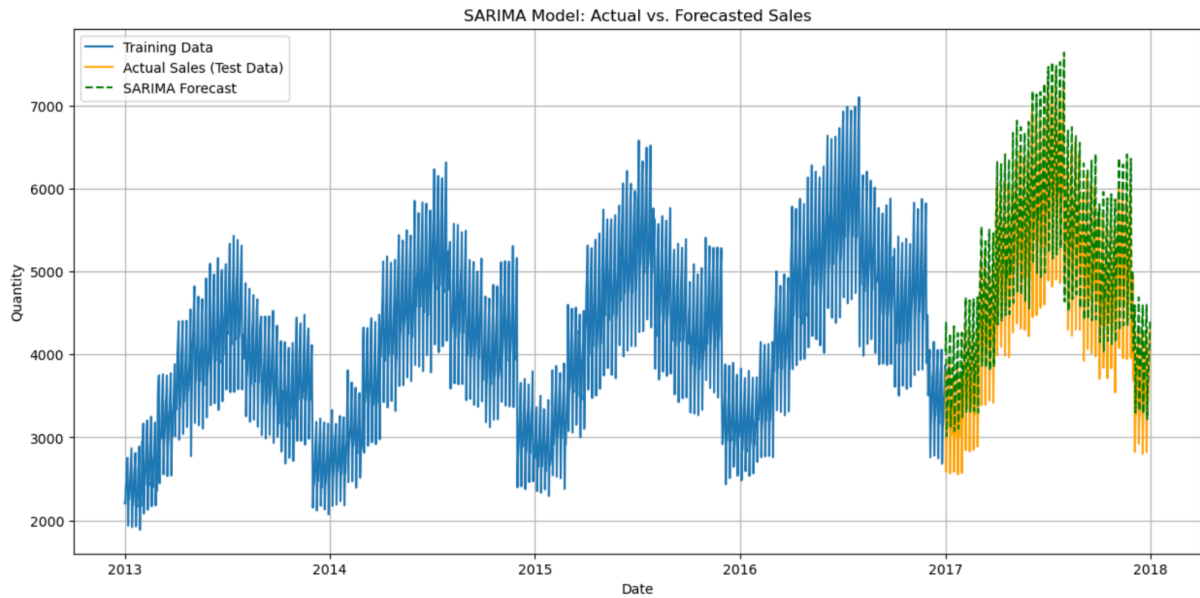
Figure 4.6: PACF Plot

4.2 Forecasting Results

Three forecasting models were developed to predict future product demand: SARIMA, Random Forest, and XGBoost. The objective was to identify the model that provides the highest forecasting accuracy while effectively capturing the temporal characteristics of retail sales data.

4.2.1 SARIMA Forecasting Results

The stationarity of the sales time series was first examined using the Augmented Dickey-Fuller (ADF) test. Since the series was found to be stationary ($p < 0.05$), no differencing was required. The SARIMA model parameters were then identified using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots before fitting the model.



4.7: Actual vs Forecasted Demand (SARIMA)

4.2.2 Random Forest Forecasting Results

Random Forest achieved improved forecasting accuracy by utilizing engineered features such as calendar variables, lag features, rolling statistics, and store-item information. The model captured nonlinear relationships more effectively than SARIMA and produced forecasts that closely followed actual sales patterns.

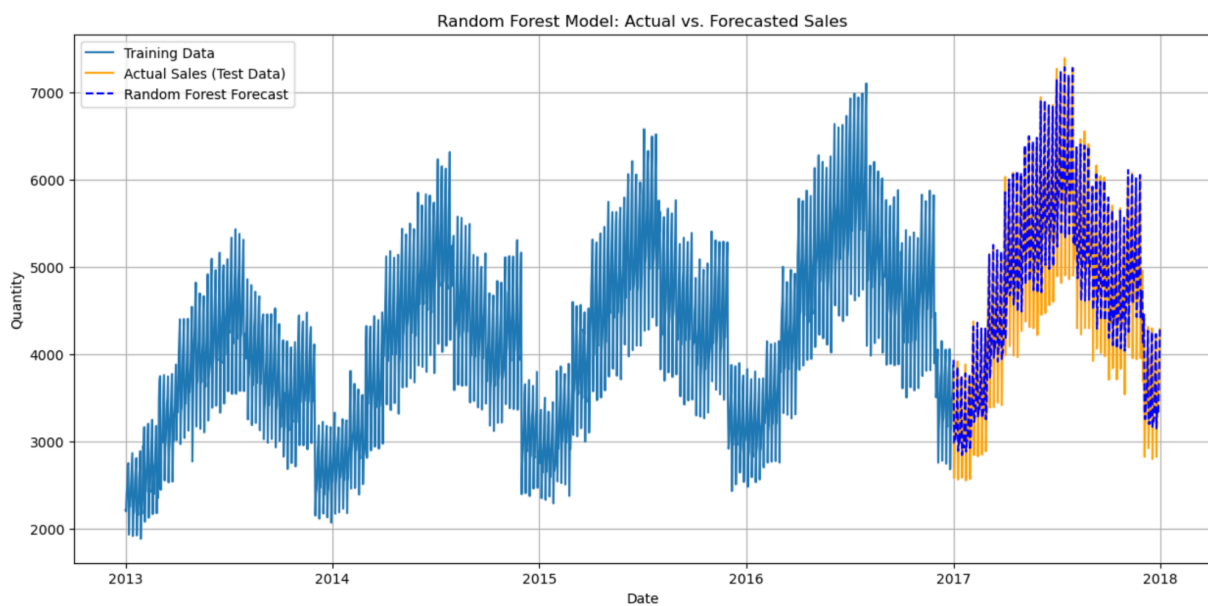


Figure 4.8: Actual vs Forecasted Demand (Random Forest)

4.2.3 XGBoost Forecasting Results

Among the evaluated models, XGBoost produced the most accurate forecasts. Its gradient boosting mechanism iteratively reduced prediction errors while effectively utilizing engineered features. The predicted values closely followed the observed sales trend and responded effectively to short-term demand fluctuations.

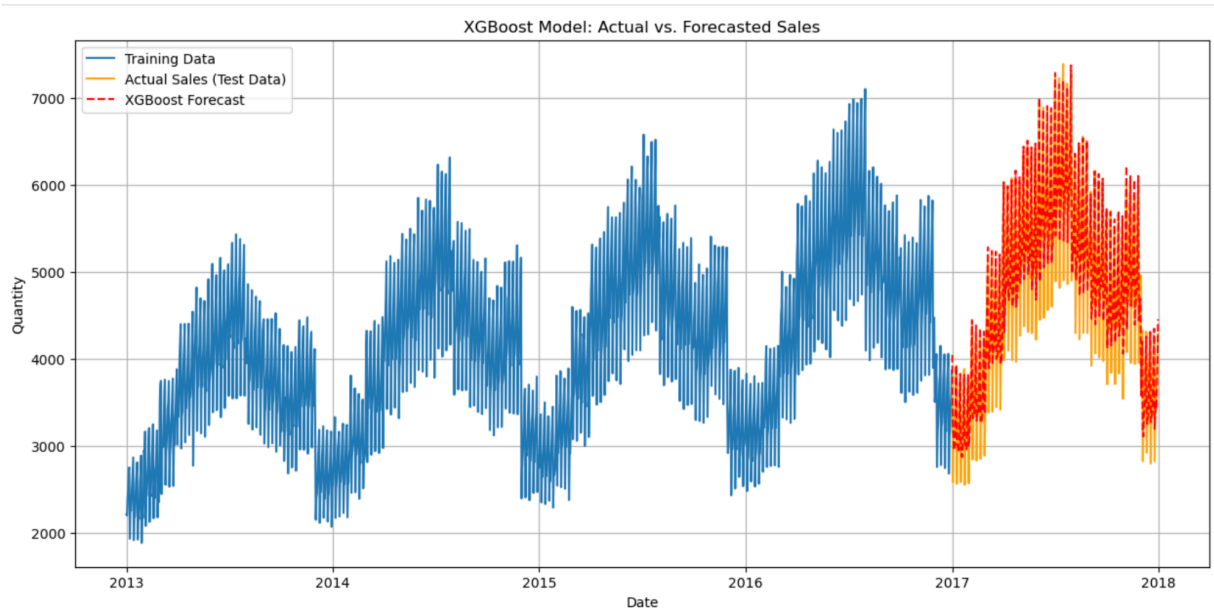


Figure 4.9: Actual vs Forecasted Demand (XGBoost)

4.3 Model Performance Comparison

The forecasting models were evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Weighted Absolute Percentage Error (WAPE), and other standard forecasting metrics.

Model Performance Comparison:

	MSE	RMSE	WAPE (%)
Model			
Random Forest	77.223075	8.787666	6.984316
XGBoost	80.883095	8.993503	7.227728
SARIMAX	765908.969914	875.162253	9.921081

Table 4.2: Model Performance Comparison

Based on the evaluation results, XGBoost achieved the highest forecasting accuracy with the lowest prediction errors. Random Forest also performed well and substantially outperformed SARIMA. Consequently, XGBoost was selected as the final forecasting model for inventory optimization.

4.4 Inventory Optimization Results

The predicted demand obtained from the XGBoost forecasting model was used to calculate EOQ, Safety Stock, and Reorder Point for the selected products.

EOQ calculations determined the optimal order quantity that minimizes the combined ordering and holding costs. Products with higher forecasted annual demand generally required larger replenishment quantities.

Safety Stock provided an additional inventory buffer against demand uncertainty and lead-time variability. Products exhibiting greater sales fluctuations required higher safety stock levels.

Reorder Point calculations combined average forecasted demand during lead time with Safety Stock, identifying the inventory level at which replenishment should begin.

Holding Cost per Unit per Year: \$10.00

--- Calculating Inventory Optimization Metrics ---

Inventory Optimization Metrics per Store-Item Combination (based on XGBoost Forecast):

	store	item_name	average_daily_forecasted_demand	annual_demand	EOQ	ROP	safety_stock	lead_time
0	1.0	13.0	55	20075	634	474	89	7
1	1.0	15.0	53	19345	622	466	95	7
2	1.0	18.0	61	22265	667	519	92	7
3	1.0	25.0	56	20440	639	478	86	7
4	1.0	28.0	64	23360	684	541	93	7
5	2.0	13.0	107	39055	884	873	124	7
6	2.0	15.0	105	38325	875	864	129	7
7	2.0	18.0	93	33945	824	775	124	7
8	2.0	25.0	111	40515	900	898	121	7
9	2.0	28.0	111	40515	900	905	128	7
10	3.0	13.0	99	36135	850	805	112	7
11	3.0	15.0	95	34675	833	782	117	7
12	3.0	18.0	99	36135	850	804	111	7
13	3.0	25.0	82	29930	774	679	105	7
14	3.0	28.0	89	32485	806	737	114	7
15	4.0	13.0	102	37230	863	817	103	7

Table 4.3: Inventory Optimization Results

These results demonstrate how demand forecasting can be translated into actionable inventory policies rather than relying solely on historical averages.

4.6 Business Discussion

The integration of machine learning-based demand forecasting with classical inventory optimization provides several practical benefits for retail organizations.

- Reduces uncertainty in procurement planning.
- Minimizes excess inventory and holding costs.
- Reduces the likelihood of stockouts.
- Improves product availability and customer satisfaction.
- Supports timely replenishment decisions.
- Enables data-driven inventory management.

Although inventory cost parameters were assumed because they were unavailable in the Kaggle dataset, the methodology can be directly implemented in real organizations by replacing the assumed parameters with actual business costs.

4.7 Limitations

The study has several limitations:

- The dataset is publicly available and may not fully represent a specific organization's operations.
- Inventory cost parameters and lead times were assumed rather than obtained from a real company.
- External factors such as promotions, weather, pricing, and economic indicators were not included.
- Only three forecasting models (SARIMA, Random Forest, and XGBoost) were evaluated.
- The inventory optimization framework assumes relatively stable lead time and inventory costs.

Despite these limitations, the proposed framework successfully demonstrates how historical sales data can be transformed into actionable inventory decisions through the integration of demand forecasting and inventory optimization.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This study presented a data-driven framework that integrates demand forecasting with inventory optimization to support effective inventory management in retail businesses. The primary objective was to develop an accurate demand forecasting model using machine learning techniques and utilize the forecasted demand to optimize inventory decisions through Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP).

This study developed a data-driven framework that integrates demand forecasting with inventory optimization to improve inventory management in retail businesses. Historical sales data were analyzed through data preprocessing, exploratory data analysis, and feature engineering before developing forecasting models using SARIMA, Random Forest, and XGBoost.

The forecasting models were evaluated using MSE, RMSE and WAPE Score. Based on the evaluation results, **XGBoost achieved the best forecasting performance** and was selected for predicting future demand. The forecasted demand was then used to calculate Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Point (ROP), providing practical inventory recommendations for efficient replenishment planning.

The proposed framework demonstrates that integrating machine learning-based demand forecasting with inventory optimization can improve forecasting accuracy, reduce inventory costs, enhance product availability, and support effective business decision-making.

5.3 Future Scope

Future research can improve forecasting accuracy by incorporating additional factors such as promotions, pricing, holidays, weather conditions, and economic indicators. Advanced deep learning models such as LSTM, GRU, and Transformer-based architectures may also be explored and compared with XGBoost. Furthermore, replacing the assumed inventory parameters with actual organizational data and integrating the framework with ERP systems or real-time inventory management platforms would enhance its applicability in real-world business environments.

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APPENDIX

Github repository link:

<https://github.com/shadhavk/demand-forecasting-and-inventory-optimization>